1. **Move All Zeros to the End**

   Problem statement:

   Given an array of N integers, move all zeros to the end while maintaining the order of non-zero elements.

   Sample Input

   text

   6

   0 1 0 3 12 0

   Sample Output

   text

   1 3 12 0 0 0

Additional test cases

Input	Expected output
5 / 1 2 3 4 5	1 2 3 4 5
5 / 0 0 1 2 3	1 2 3 0 0
4 / 0 0 0 0	0 0 0 0

C Answer Code

c

```
#include <stdio.h>
int main()
{
    int n, a[100], i, j = 0;
    scanf("%d", &n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    for(i = 0; i < n; i++)
    {
        if(a[i] != 0)
            a[j++] = a[i];
    }
    while(j < n)
        a[j++] = 0;
```

```
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}
```

2. Check Whether Two Strings Are Anagrams

Problem statement:

Two strings are anagrams if they contain the same characters with the same frequencies. Check whether two given strings are anagrams. Assume lowercase English letters.

Sample Input

text
listen
Silent

Sample Output

text
Anagram

Additional test cases

Input	Expected output
`race / care`	`Anagram`
`hello / world`	`Not Anagram`
`abc / abcd`	`Not Anagram`
`aabb / bbaa`	`Anagram`

C Answer Code

c

```
#include <stdio.h>
#include <string.h>
int main()
{
```

```
    char a[100], b[100];
    int count[26] = {0};
    int i;
    scanf("%99s", a);
    scanf("%99s", b);
    if(strlen(a) != strlen(b))
    {
        printf("Not Anagram");
        return 0;
    }
    for(i = 0; a[i] != '\0'; i++)
    {
        count[a[i] - 'a']++;
        count[b[i] - 'a']--;
    }
    for(i = 0; i < 26; i++)
    {
        if(count[i] != 0)
        {
            printf("Not Anagram");
            return 0;
        }
    }

    printf("Anagram");

    return 0;
}
```

3. Rotate an Array by K Positions

Problem statement:

Given an array of N integers and an integer K, rotate the array to the right by K positions.

Sample Input

text
5
1 2 3 4 5
2

Sample Output

text
4 5 1 2 3

Additional test cases

Input	Expected output
4 / 1 2 3 4 / 1	4 1 2 3
3 / 1 2 3 / 3	1 2 3
3 / 1 2 3 / 5	2 3 1

C Answer Code

```
c
#include <stdio.h>
int main()
{
    int n, a[100], temp[100];
    int k, i;
    scanf("%d", &n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    scanf("%d", &k);
    k = k % n;
    for(i = 0; i < n; i++)
        temp[(i + k) % n] = a[i];
    for(i = 0; i < n; i++)
        printf("%d ", temp[i]);
    return 0;
}
```